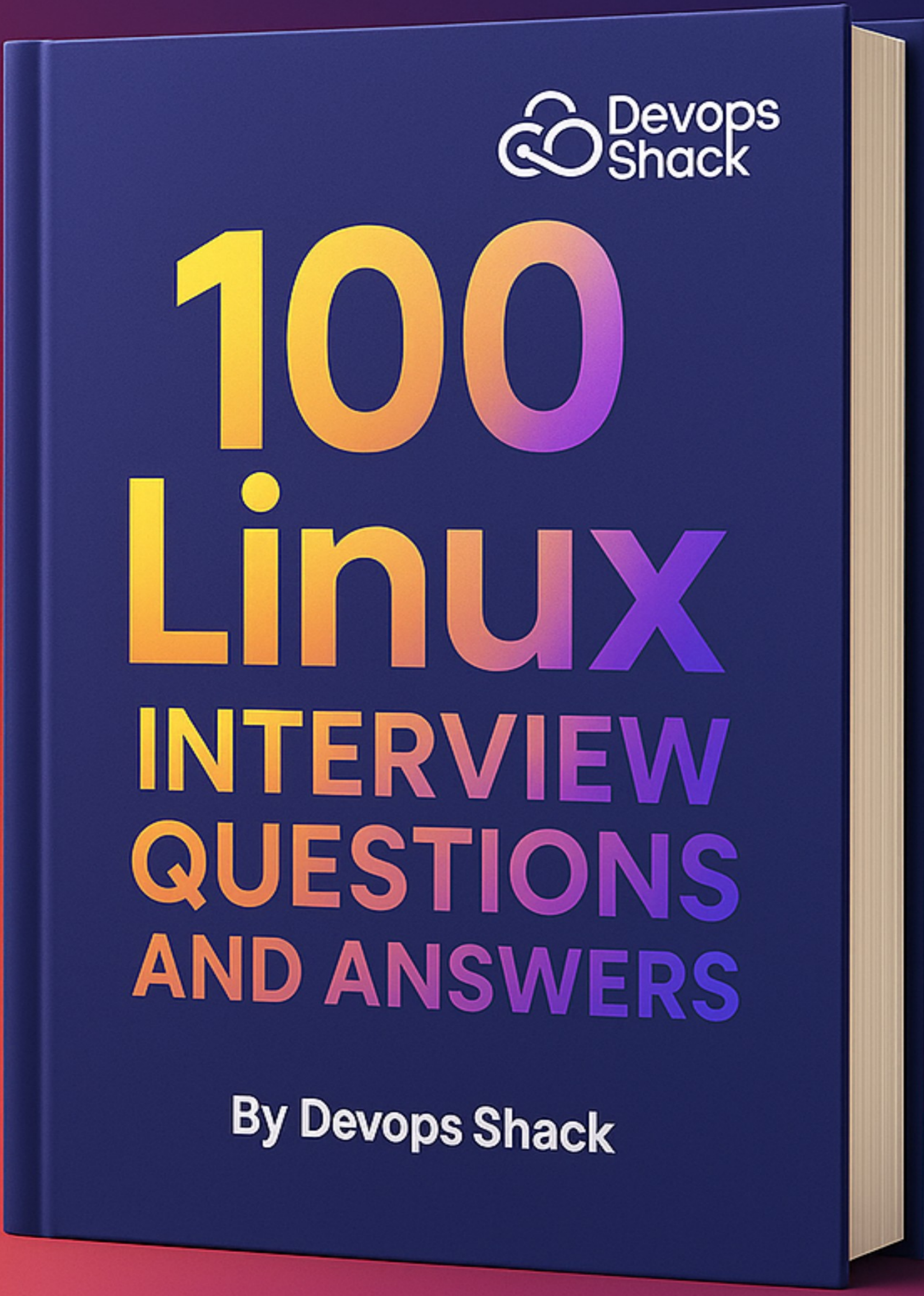
Devops Shack
100 Linux
INTERVIEW QUESTIONS AND ANSWERS
By Devops Shack

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100 Linux Interview Questions and Answers

1. Question: What is the Linux file system hierarchy?

Answer: Linux organizes everything under a single unified tree that starts with `/`. System configurations live in `/etc`, logs in `/var`, user files in `/home`, and applications in `/usr`. This standardized structure ensures predictable automation across all distributions.

2. Question: What do r, w, x permissions mean?

Answer: `r` allows reading a file or listing a directory, `w` allows editing/deleting content, and `x` allows executing a file or entering a directory. These permissions apply to the owner, group, and others. This structure enforces controlled access in multi-user environments.

3. Question: How are chmod, chown, and chgrp different?

Answer: `chmod` modifies file permissions, `chown` changes the file owner and group, and `chgrp` changes only the group ownership. They work together to secure files and properly allocate access in shared environments.

4. Question: What is umask?

Answer: Umask defines the default permissions for new files and directories by masking certain bits. For example, `umask 022` results in 644-permissions for files and 755 for directories. It prevents insecure defaults when applications create files.

5. Question: What are SUID, SGID, and Sticky Bit?

Answer: SUID lets a program run with the file owner's privileges, commonly used in programs like `passwd`. SGID makes new files inherit the directory's group. Sticky Bit restricts deletion inside shared folders so only file owners can delete their own files.

6. Question: How does networking work in Linux?

Answer: Traffic flows from applications → sockets → transport layer (TCP/UDP) → IP routing → Ethernet → NIC. Linux provides tools like `ip`, `ss`, `ethtool`, and `iptables/nftables` to manage interfaces, ports, and packet filtering.

7. Question: What is the difference between TCP and UDP?

Answer: TCP is connection-oriented, reliable, ordered, and slower — ideal for SSH, HTTP, and Git. UDP is connectionless, fast, lightweight, and unreliable — ideal for DNS, voice/video streaming, and gaming.

8. Question: Which commands are most useful for network troubleshooting?

Answer: `ip a` examines network interfaces, `ss -tulpn` shows listening ports, `ping` tests connectivity, `curl` tests HTTP/S access, `dig/nslookup` troubleshoot DNS, and `traceroute/mtr` trace routing paths. These tools help isolate failures quickly.

9. Question: What is the purpose of /etc/hosts and /etc/resolv.conf?

Answer: `/etc/hosts` provides local static hostname-to-IP mapping, especially useful in offline or internal environments. `/etc/resolv.conf` specifies DNS servers for hostname resolution. Together, they control how Linux resolves names.

10. Question: What is ARP in Linux networking?

Answer: ARP translates an IPv4 address to a MAC address so packets can travel within the LAN. Linux stores these mappings in an ARP cache visible via `ip neigh` or `arp -a`. Incorrect ARP entries can cause connectivity issues.

11. Question: What is systemd?

Answer: systemd is the init and service manager responsible for booting the system, starting services, managing logs, and handling dependencies. It replaced SysVinit to support fast booting, parallel service startup, and modern automation.

12. Question: What is a systemd service unit file?

Answer: It defines how a service should run — including command path, user, restart policy, dependencies, and boot targets. Unit files allow services to behave consistently across restarts and reboots, improving reliability.

13. Question: What are the most common systemctl commands?

Answer: `systemctl start/stop/restart` manages service lifecycle, `status` checks health, `enable/disable` configures auto-start, and `daemon-reload` reloads systemd after editing a service file. These are essential in Linux troubleshooting.

14. Question: Difference between systemctl and journalctl?

Answer: systemctl manages services (start, stop, enable) while journalctl retrieves logs recorded by systemd. Together they provide complete visibility into service status and failure reasons.

15. Question: Where are Linux system logs stored?

Answer: System logs live under `/var/log/`, including `syslog/messages` for general system events, `auth.log/secure` for authentication, and `kern.log` for kernel messages. Logs are the first place to check when troubleshooting production servers.

16. Question: What is log rotation?

Answer: Log rotation prevents log files from filling storage by automatically rotating, compressing, and deleting old logs. It is configured with `logrotate` and helps maintain long-term observability without performance issues.

17. Question: Difference between apt, yum/dnf, and zypper?

Answer: apt is the package manager for Debian/Ubuntu, yum/dnf for RHEL/Amazon Linux, and zypper for SUSE. All support installing, updating, removing packages, and handling dependencies from repositories.

18. Question: What is a package repository?

Answer: A repository is a server hosting installable packages and metadata. Package managers communicate with repositories to fetch, update, and verify software. Updates and version control depend on repository configuration.

19. Question: What are dpkg and rpm?

Answer: dpkg installs `.deb` packages and rpm installs `.rpm` packages locally without automatic dependency resolution. They are low-level installers beneath apt and yum/dnf respectively.

20. Question: What are Linux process states?

Answer: The most common states are running, sleeping (waiting), stopped, uninterruptible wait (I/O), and zombie (terminated but not reaped). Understanding these states helps identify performance and crash issues.

21. Question: What's the difference between a process and a thread?

Answer: A process has its own independent memory space, while threads share memory inside the same process. Threads are lightweight and faster, which is why they're preferred for parallel computing.

22. Question: What are PID and PPID?

Answer: PID is a unique identifier assigned to every process. PPID is the parent process ID, representing the process that created it. Tracking PID & PPID helps spot orphaned or runaway processes.

23. Question: What is a zombie process?

Answer: A zombie is a terminated process that still exists in the process table because its parent hasn't collected its exit status. It consumes no memory/CPU, but many zombies indicate a faulty parent application.

24. Question: What are nice and renice?

Answer: `nice` starts a process with a defined CPU priority, while `renice` changes the priority of an already running process. Lower values mean higher priority and can influence scheduling fairness on busy servers.

25. Question: What is SELinux/AppArmor?

Answer: These are mandatory access control security systems that restrict application capabilities beyond standard file permissions. They limit damage in case of service compromise and are widely used in production environments.

26. Question: How do you check disk usage in Linux?

Answer: `df -h` shows filesystem-level space usage, while `du -sh *` shows space consumed by each directory. These commands help identify whether storage issues are global or caused by a specific folder. For deeper analysis, tools like `ncdu` visually highlight large files.

27. Question: What is swap memory?

Answer: Swap is disk space used when RAM becomes full. It prevents the system from crashing under memory pressure but is slower than physical RAM. Excessive swap usage indicates memory constraint or poorly coded applications.

28. Question: What is the OOM killer?

Answer: The kernel's Out-Of-Memory killer terminates the most memory-hungry process during low-memory situations. It prevents total system freeze. Frequent OOM events usually indicate insufficient RAM or memory leaks.

29. Question: What is the difference between Hard Link and Soft Link?

Answer: Hard links point directly to the file's inode, so they continue to work even if the original file is deleted. Soft links (symlinks) point to the file path and break if the original is removed. Soft links are used more commonly for redirection.

30. Question: What is ulimit in Linux?

Answer: `ulimit` controls resource consumption per user or process, such as number of open files, maximum process count, or stack size. It prevents a single faulty application from consuming all system resources.

31. Question: What is /proc filesystem?

Answer: `/proc` is a virtual filesystem that exposes kernel and process runtime information. It includes details like CPU, memory, kernel parameters, and per-process directories. Nothing inside `/proc` is physically stored on disk.

32. Question: What is the purpose of /etc/fstab?

Answer: `/etc/fstab` contains filesystem mount instructions that run automatically during boot. It defines device, mount point, filesystem type, and mount options. Mistakes in fstab can prevent the OS from booting.

33. Question: What is tmpfs?

Answer: `tmpfs` is a temporary filesystem stored in RAM to provide extremely fast read/write access. It's often used for runtime files like cache, PID files, and temporary storage. Its contents disappear on reboot.

34. Question: What is I/O wait?

Answer: I/O wait is the time the CPU sits idle waiting for slow disk operations to complete. High I/O wait typically signals disk bottlenecks or overloaded filesystems. It can be monitored using `top`, `vmstat`, or `iostat`.

35. Question: What does lsof do?

Answer: `lsof` lists open files — including regular files, sockets, and network ports. It is useful for diagnosing which process is using a port or holding a file that can't be deleted because it's "in use."

36. Question: What are system calls?

Answer: System calls are low-level functions used by applications to request kernel operations like file access, network communication, and memory allocation. They enable controlled interaction between user space and kernel space.

37. Question: What is the difference between Kernel Space and User Space?

Answer: Kernel space runs core OS services with full privilege, while user space runs applications with restricted permissions. This separation prevents applications from crashing the OS and increases security.

38. Question: What are cgroups in Linux?

Answer: cgroups (Control Groups) limit and monitor resource usage — CPU, memory, and I/O — per process or container. They are fundamental to container platforms like Docker and Kubernetes for resource isolation.

39. Question: What are namespaces in Linux?

Answer: Namespaces isolate system resources like process IDs, networks, mount points, hostnames, and users. They allow multiple independent environments to operate on the same host — forming the foundation of containers.

40. Question: How do you check which services are listening on ports?

Answer: `ss -ltnp` displays listening TCP ports with the associated process ID, while `lsof -i :<port>` shows which application is using a specific port. These commands help troubleshoot "port already in use" errors.

41. Question: What is the difference between Cron and Systemd timers?

Answer: Cron provides basic time-based scheduling with minimal logging. Systemd timers support logging, service dependencies, boot-time triggers, and retry policies. Timers are more reliable for production automation.

42. Question: What is the /etc/sudoers file used for?

Answer: It defines which users or groups can perform privileged actions using `sudo`. The file is edited through `visudo` to prevent syntax errors that could lock out administrators.

43. Question: What are passwd, shadow, and group files?

Answer: `/etc/passwd` stores user account details (not passwords), `/etc/shadow` stores encrypted password hashes, and `/etc/group` stores group membership. These files form the foundation of Linux local authentication.

44. Question: What are kernel tuning parameters?

Answer: Kernel tuning parameters, modified using `sysctl` or `/etc/sysctl.conf`, adjust system performance — for example, network throughput, file handle limits, or memory swapping. They are heavily used in high-load environments.

45. Question: How do you check processes running on a port?

Answer: `ss -ltnp` shows the process ID and service tied to a listening port. Alternatively, `lsof -i :<port>` identifies exactly which application is occupying the port — critical for diagnosing deployment conflicts.

46. Question: How does Linux resolve DNS queries?

Answer: The resolver follows a lookup order defined in `/etc/nsswitch.conf`: it checks `/etc/hosts` first, then DNS servers defined in `/etc/resolv.conf`. This design allows offline hostname mapping and external resolution.

47. Question: What is LDAP authentication on Linux?

Answer: LDAP authentication offloads user identity and password validation to a centralized directory like Active Directory. It allows single sign-on and eliminates local user management across multiple servers.

48. Question: What is journald log persistence?

Answer: By default, journald logs are stored in memory and lost after reboot. Creating `/var/log/journal/` enables persistent logging, helping in long-term troubleshooting and auditing.

49. Question: How does Linux handle signals like SIGKILL and SIGTERM?

Answer: SIGTERM requests a graceful shutdown so the process can save state, while SIGKILL forces immediate termination and cannot be caught or ignored. Signal handling is key to safe service shutdown.

50. Question: What is a solid troubleshooting methodology on Linux?

Answer: Start by checking logs, then verify service status, examine CPU/memory/disk metrics, test network connectivity, and check permissions/SELinux. Fixing Linux issues is about correlating symptoms rather than guessing.

51. Question: What is GRUB in Linux?

Answer: GRUB (Grand Unified Bootloader) is the program responsible for loading the Linux kernel at system startup. It provides a boot menu and allows selecting different kernels or OS installations. Without GRUB or another bootloader, the system cannot start.

52. Question: What is initrd / initramfs?

Answer: initrd is a temporary root filesystem loaded into memory before the real root filesystem mounts. It contains essential drivers needed during boot, such as storage and filesystem drivers. Without initrd, the kernel may not access the root filesystem.

53. Question: What are systemd targets (previously runlevels)?

Answer: Targets define system boot states such as multi-user mode, graphical mode, rescue mode, and emergency mode. They group related services together, allowing fast and parallel booting. `systemctl get-default` shows the current default target.

54. Question: What is rescue mode in Linux?

Answer: Rescue mode boots into a minimal environment with only critical services and no multi-user networking. It’s used to fix issues like boot failures, fstab errors, or forgotten root passwords. It avoids loading unnecessary daemons.

55. Question: Difference between `reboot` and `systemctl reboot`?

Answer: Both restart the system, but `systemctl reboot` uses systemd to gracefully stop services following dependencies before restart. It’s recommended for production machines to avoid abrupt shutdown.

56. Question: What is hostnamectl used for?

Answer: `hostnamectl` displays system identity details and updates the hostname persistently. It supports static, pretty, and transient hostnames. It ensures hostname survives across reboots.

57. Question: What is timedatectl?

Answer: `timedatectl` configures system timezone, NTP synchronization, and date/time settings. Correct NTP sync prevents drift issues, which are critical for certificates, logs, and distributed systems.

58. Question: What is chroot and why is it used?

Answer: `chroot` changes the root filesystem of a running process. It isolates the process in a controlled environment useful for system recovery, password reset, or building minimal containers.

59. Question: What is strace used for?

Answer: `strace` traces system calls made by a process, helping debug issues like missing files, permission failures, or unexpected termination. It's extremely valuable when logs don't show the failure reason.

60. Question: What is tcpdump, and when do you use it?

Answer: `tcpdump` captures and displays network packets in real time. It's used to analyze connectivity problems, DNS issues, slow applications, or unauthorized traffic. It's the first tool in network packet-level debugging.

61. Question: What does traceroute do?

Answer: `traceroute` displays the path packets travel from source to destination across routers. It helps identify latency or hops causing slowdown or drop. Useful for debugging WAN or multi-network routing.

62. Question: What is mtr and how is it different from traceroute?

Answer: `mtr` combines ping and traceroute into a live dashboard showing packet loss and latency per hop. Unlike traceroute's snapshot, mtr provides real-time network health visibility.

63. Question: What does the ping command test?

Answer: `ping` checks host reachability and reports round-trip time and packet loss. It verifies basic connectivity but doesn't guarantee higher-layer protocols like HTTP are functioning.

64. Question: What is network bonding / NIC teaming?

Answer: Bonding aggregates multiple network interfaces to improve bandwidth and/or redundancy. If one NIC fails, traffic automatically switches to another. It is common in critical production servers.

65. Question: What is a routing table?

Answer: A routing table decides where outgoing packets should be delivered. It tracks subnets, gateways, and interfaces, and is viewed using `ip r`. Incorrect routes cause unreachable networks.

66. Question: Static routing vs dynamic routing in Linux?

Answer: Static routing requires admin-defined routes and works best in stable networks. Dynamic routing learns routes automatically using protocols like BGP or OSPF, enabling scalable routing.

67. Question: What is /etc/sysconfig/network-scripts used for?

Answer: In RHEL/CentOS, this directory stores network interface config files like `ifcfg-eth0`. They define IP, gateway, DNS, and routing options. Editing them triggers persistent network settings across reboots.

68. Question: What is a hostname in Linux?

Answer: A hostname uniquely identifies a machine on a network. It's used in logs, SSH prompts, and hostname resolution. It can be set temporarily or permanently using `hostnamectl`.

69. Question: Purpose of ssh-copy-id?

Answer: `ssh-copy-id` installs your public key on a remote machine so you can log in without a password. It automates passwordless authentication setup for remote access and DevOps automation.

70. Question: What is the known_hosts file?

Answer: `known_hosts` stores SSH fingerprints of previously connected servers. It helps prevent man-in-the-middle attacks by verifying remote server authenticity in future connections.

71. Question: What is the difference between /bin and /usr/bin?

Answer: `/bin` contains essential binaries required for boot and repair operations, while `/usr/bin` contains regular user applications. `/bin` is sometimes available even in recovery shells.

72. Question: What is /dev directory?

Answer: `/dev` holds device files representing storage devices, partitions, RAM, serial ports, and pseudo-devices like `/dev/null`. Programs interact with hardware via these files.

73. Question: What is /run directory?

Answer: `/run` stores volatile runtime data such as PID files and sockets. It's tmpfs-based and cleared after reboot. Services rely on `/run` to track their current state.

74. Question: What is /mnt vs /media?

Answer: `/mnt` is used for temporary manual mount points, while `/media` is automatically used by the OS to mount removable drives like USB and CD/DVD. This helps organize external storage consistently.

75. Question: journald vs rsyslog — what's the difference?

Answer: journald stores logs in a binary journal with indexed search via `journalctl`, while rsyslog writes logs to text files in `/var/log`. Many distributions run both — journald collects logs and forwards them to rsyslog.

76. Question: What is a kernel panic?

Answer: A kernel panic occurs when the Linux kernel encounters a fatal condition and cannot continue safely. It freezes the system to avoid data corruption. Common causes include faulty drivers, hardware errors, and filesystem corruption.

77. Question: What is a core dump?

Answer: A core dump is a file containing the memory contents of a crashed program at the moment of failure. It helps developers debug segmentation faults, memory issues, and unhandled exceptions. Core dumps need `ulimit -c` enabled to be generated.

78. Question: What is a memory leak in Linux?

Answer: A memory leak happens when a process keeps allocating memory without releasing it back to the OS. Over time, the system slows down and may trigger the OOM killer. Tools like `top`, `ps`, and `/proc/<PID>/smaps` help detect leaks.

79. Question: What is load average in Linux?

Answer: Load average represents the average number of runnable or waiting tasks over 1, 5, and 15 minutes. It indicates system workload relative to CPU capacity. If load > CPU cores, it suggests performance pressure.

80. Question: What is the ps command used for?

Answer: `ps` lists running processes and resource usage. The common form `ps aux` shows CPU, memory usage, start time, and command. It is often combined with `grep` for targeted process checks.

81. Question: What is the difference between kill and pkill?

Answer: `kill` terminates a process using its PID, while `pkill` terminates processes by name or pattern. `pkill` is convenient when the PID is unknown but caution is needed to avoid killing multiple services unintentionally.

82. Question: What is nohup used for?

Answer: `nohup` allows programs to continue running even after closing the terminal. It redirects output to `nohup.out` by default and is commonly used for long-running scripts and background jobs.

83. Question: What are screen and tmux?

Answer: `screen` and `tmux` provide persistent terminal sessions that stay alive even after SSH disconnects. They allow multiple windows, session sharing, and job recovery — essential for remote server administration.

84. Question: What are cron.allow and cron.deny?

Answer: These files control which users can use cron. If `cron.allow` exists, only users listed inside can schedule cron jobs. If not, users listed in `cron.deny` are blocked. It helps enforce automation governance.

85. Question: What is stored inside /var/spool?

Answer: `/var/spool` holds temporary queued tasks such as print jobs, cron jobs, and mail messages waiting to be processed. It acts as a staging area for scheduled or deferred tasks.

86. Question: What is rsync and why is it used?

Answer: rsync is a fast differential file transfer tool that synchronizes only changed blocks, not entire files. It's widely used for backups, deployments, and incremental sync between servers.

87. Question: What is SSH port forwarding?

Answer: SSH forwarding securely tunnels TCP traffic through an SSH connection. It allows access to private/internal services without exposing them publicly over the internet. Common types are local, remote, and dynamic forwarding.

88. Question: What is a filesystem UUID?

Answer: A UUID uniquely identifies storage partitions regardless of device names like /dev/sda1. Using UUIDs in /etc/fstab avoids mount failures if device names change due to hardware reordering.

89. Question: Difference between ext4, XFS, and Btrfs?

Answer: ext4 is stable and widely supported, XFS is optimized for large files and high throughput, and Btrfs supports advanced features like snapshots and built-in RAID. Selection depends on storage requirements.

90. Question: What is an inode?

Answer: An inode stores metadata about a file — size, timestamps, permissions, and data block pointers — but not the filename itself. Filenames map to inodes via directories.

91. Question: Difference between df and du commands?

Answer: df shows disk usage per filesystem, while du shows disk usage at directory/file level. High df usage but low du usage may indicate deleted files still held open by a process.

92. Question: Purpose of mount and umount?

Answer: mount attaches storage devices or partitions to the Linux directory tree at a mount point. umount safely detaches them, ensuring no pending reads/writes remain to prevent corruption.

93. Question: What does lsblk do?

Answer: lsblk lists block devices and their relationships like disk → partition → filesystem. It's an excellent tool to visualize storage layout without modifying anything.

94. Question: What is fdisk used for?

Answer: fdisk is used to create and modify partitions on disks. It supports listing existing partitions, resizing, deleting, and generating partition tables. It requires care to avoid data loss.

95. Question: What is partprobe?

Answer: partprobe informs the kernel that partition tables have changed without requiring a reboot. This is useful after using fdisk or storage resize tools.

96. Question: What is fstrim and why is it important?

Answer: `fstrim` notifies SSDs of unused blocks so they can be reclaimed. It improves SSD performance and lifespan. Most modern systems schedule trim weekly or daily via systemd.

97. Question: What is the history command used for?

Answer: `history` displays previously executed shell commands. It's useful for audit, debugging, and quickly re-executing past commands. It is stored per user in `~/.bash_history`.

98. Question: What is the /etc/skel directory?

Answer: `/etc/skel` contains default configuration files that are copied to a new user's home directory during account creation. It provides a standardized default environment.

99. Question: What is an SSH key pair?

Answer: An SSH key pair consists of a private key stored locally and a public key stored on the server. It enables secure passwordless authentication and stronger security than passwords.

100. Question: What makes a Linux engineer strong for DevOps roles?

Answer: A strong DevOps Linux engineer understands OS internals, networking, systemd services, logs, package management, and performance troubleshooting. They don't just run commands — they can diagnose and fix root causes under pressure.